

Retail & Supply Chain Analytics

15 Advanced SQL Practice Questions (No Solutions)

Module 1: The Basics & Time Series

1. Year-over-Year (YoY) Sales & Profit Growth

Har saal humari sales aur profit kaise badhe? Year-wise breakdown nikalen.

2. Seasonality (The Weekend Effect)

Hafta ke kis din (Day Name) sabse zyada sales hoti hai aur kis din sabse zyada returns aate hain?

3. The Best Quarter

Ab tak ka sabse behtareen quarter (e.g., 2017 - Q4) kaun sa raha hai jisme highest sales aayi?

Module 2: Profit Bleeding (Where are we losing money?)

4. The Discount Trap

Aisi kaun si category aur sub-category hain jahan discount dene ke chakkar mein bhari nuksan (negative profit) hua?

5. The Return Problem

Kis Region mein customers sabse zyada products return karte hain? Return rate (%) calculate karen.

6. Top 10 Loss-Making Customers

Wo 10 customers kaun hain jinhe serve karne mein company ne overall sabse zyada paisa gavaya hai?

Module 3: Advanced Supply Chain & Logistics

7. Shipping Delay Analysis

Alag-alag ship mode (Standard, First Class, etc.) mein delivery mein ausatan (average) kitne din lagte hain?

8. Late Delivery Impact

Kya late delivery (3 din se zyada) hone par customers product zyada return karte hain? (Compare On-Time vs Late Returns)

9. Salesperson Performance

Kaun sa salesperson sabse zyada sale la raha hai aur kiska profit margin (%) sabse kharab hai?

Module 4: The "WOW" Factors (Window Functions & CTEs)

10. The Pareto Principle (80/20 Rule)

Kya humare top customers humari 80% sale laate hain? (Cumulative Sum ka use karke top customers ka % nikalen).

11. Customer Churn

Wo customers find karen jinhone 2015 ya 2016 mein shopping ki thi, lekin 2017 mein ek bhi order nahi diya.

12. 30-Day Moving Average

Sale ke trend ko smooth karne ke liye daily sales ka 30-day "Rolling Average" calculate karen.

13. Customer Segmentation (Mini RFM)

Customers ko unke order count aur total sales ke hisab se 'VIP', 'Regular' aur 'One-Time Buyer' segments mein bantein.

14. Month-over-Month (MoM) Growth

Pichle mahine ke mukable is mahine humari sale kitne % badhi ya ghati? (Use LAG function).

15. Most Profitable Route

Kis State aur City ke combination mein per-order sabse zyada profit hota hai? (Minimum 10 orders ki condition ke sath).
